Nesterov Acceleration and Point Convergence: Context for a Recent Proof

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1. Introduction

The minimisation of smooth, convex functions by an iterative, first-order method at an $O(1/k)$ rate is a well-known result, with origins dating back to Cauchy [1]. The key device, known as *gradient descent*, has several desirable properties, including so-called *point convergence*—that the iterates themselves converge as the number of iterations increases indefinitely. A break-through in the understanding of the method occurred when it was established that the fastest theoretical rate for a first-order algorithm on this *class* of problem was $O(1/k^2)$. For some time, it was unknown whether such an algorithm attaining this maximum rate existed.

The answer eventually came in the affirmative from Nesterov's seminal paper [2], in which it was proved that the method could be accelerated to $O(1/k^2)$ with almost no additional computational burden. In the decades since, a large and diverse body of results on related problems has been established as optimisation has taken on renewed relevance in applications, with Artificial Intelligence being the dominant contemporary example. However, whether point convergence was retained under Nesterov's acceleration scheme remained an open question until very recently, when researchers were again able to answer in the affirmative [3]. Perhaps fittingly, the the key innovation that provided sthe solution this longstanding open question was obtained with the assistance of a large language model.

The rest of this work traces a narrow path through the literature, with the goal of providing a tightly-focused context for recent proof (Theorem 3.5 in [3]) in terms of the preceeding literature.

2. Minimisation of smooth, convex functions

2.1. Problem of interest, key concepts

In this work, the pricipal problem of interest is the minimisation of real-valued, smooth, and convex functions by iterative algorithms. Formally, let $n \in \mathbb{N}^\star$ and $f : \mathbb{R}^n \to \mathbb{R}$ be a real-valued function. The (unconstrained) optimisation problem is

$$\underset{x \in \mathbb{R}^n}{\operatorname{minimise}} f(x), \tag{1}$$

where f is 'smooth' and convex. To be more precise about terms, I now establish some

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Bibliography

[1] A. Cauchy and others, “Méthode générale pour la résolution des systemes d'équations simultanées,” *Comp. Rend. Sci. Paris*, vol. 25, no. 1847, pp. 536–538, 1847.

[2] Y. E. Nesterov, “A method of solving a convex programming problem with convergence rate $O(1/k^2)$”, *Dokl. Akad. Nauk SSSR*, vol. 269, no. 3, pp. 543–547, 1983.

[3] U. Jang and E. K. Ryu, "Point Convergence of Nesterov's Accelerated Gradient Method: An AI-Assisted Proof." 2026.